Experiment - 1

1) 8 Bit addition :

* Load the address of first 8 bit number
* Put first number into AL register
* Increment si and get second number to BL
* Add second number with AL
* Store the result in memory.

CODE :

```
code segment
assume cs : code
Start :
    mov ah, 0000H
    mov si, 3000h
    mov al , [si]
    inc si
    mov bl, [si]
    add al, bl
    inc si
    mov [si], al
    jc l1
    inc si
    mov [si] , ah
    int 3
l1 : inc ah
     inc si
     mov [si] , ah
     int 3
```

code ends
end start

OUTPUT: 075A : 3000 B8.CD 06.FF

AX = 01BC BX = 00EF CX = 001B DX = 0000 SP = 0000

BP = 0000 SI = 3003 DI = 0000 DS = 075A ES = 075A

SS = 0769 CS = 076A IP = 001A

```
C:\8086>debug add8.exe
-G=0000

AX=00BE  BX=0006  CX=001B  DX=0000  SP=0000  BP=0000  SI=3003  DI=0000
DS=075A  ES=075A  SS=0769  CS=076A  IP=0014   NV UP EI PL NZ NA PE NC
076A:0014 CC            INT     3
-E3000
075A:3000  B8.CD   06.EF

-G=0000

AX=01BC  BX=00EF  CX=001B  DX=0000  SP=0000  BP=0000  SI=3003  DI=0000
DS=075A  ES=075A  SS=0769  CS=076A  IP=001A   NV UP EI PL NZ NA PE CY
076A:001A CC            INT     3
-
```

2) 8 bit Subtraction :

* Load the address of first 8 bit number
* Move first number into AL register
* Load address of second bit number by incrementing
* Subtract second number from AL
* Store result from AL.

CODE :

```
code segment
assume cs: code
```

```
start:
    mov ah, 0000h
    mov si, 3000h
    mov al, [si]
    inc si
    mov bl, [si]
    Sub al, bl
    inc si
    mov [si], al
    jc l1
    inc si
    mov [si], ah
    int 3
    l1: inc ah
        inc si
        mov [si], ah
        int 3
code ends
end start
```

OUTPUT:

075A : 3000 B8 . 2B 06 . 1A

G = 0000

AX = 0011 BX = 001A CX = 001B DX = 0000 SP = 0000

BP = 0000 SI = 3003 DI = 0000 DS = 075A ES = 075A

SS = 0769 CS = 076A IP = 0014

0/p verified

4/11/21

```
C:\8086>debug sub8.exe
-G=0000

AX=00B2  BX=0006  CX=001B  DX=0000  SP=0000  BP=0000  SI=3003  DI=0000
DS=075A  ES=075A  SS=0769  CS=076A  IP=0014   NV UP EI PL NZ NA PE NC
076A:0014 CC            INT     3
-E3000
075A:3000  B8.2B   06.1A

-G=0000

AX=0011  BX=001A  CX=001B  DX=0000  SP=0000  BP=0000  SI=3003  DI=0000
DS=075A  ES=075A  SS=0769  CS=076A  IP=0014   NV UP EI PL NZ NA PE NC
076A:0014 CC            INT     3
-
```

3) 8 bit Multiplication :

* Initialize data segment and load the initial address and get input of first number in AL
* Increment si and get second 8 bit number
* Move it to BL and multiply it with AL.
* Result stores in AX.

CODE :

```
code segment
assume cs : code
start :
    mov si, 7000h
    mov al, [si]
    inc si
    mov bl, [si]
    mul bl
    inc si
```

```
    mov [si], ax
    int 3
code ends
end start
```

OUTPUT :

075A : 7000 10. 09 F7. 02

-G=0000

AX = 0012 BX = 0002 CX = 000E DX = 0000 SP = 0000

BP = 0000 SI = 7002 DI = 0000 DS = 075A ES = 075A

SS = 0769 CS = 076A IP = 000D

3)

```
C:\8086>debug mul8.exe
-
-g=0000

AX=0F70  BX=00F7  CX=000E  DX=0000  SP=0000  BP=0000  SI=7002  DI=0000
DS=075A  ES=075A  SS=0769  CS=076A  IP=000D   NV UP EI PL NZ NA PO CY
076A:000D CC            INT     3
-e7000
075A:7000  10.09   F7.02

-g=0000

AX=0012  BX=0002  CX=000E  DX=0000  SP=0000  BP=0000  SI=7002  DI=0000
DS=075A  ES=075A  SS=0769  CS=076A  IP=000D   NV UP EI PL NZ NA PO NC
076A:000D CC            INT     3
-
```

4) 16 bit addition :

```
code segment
assume cs:code
Start:
    mov ax, 0000h
    mov bx, ax
    mov dx, ax
```

```
mov si, 3000h
mov ax, [si]
inc si
inc si
mov bx, [si]
inc si
inc si
add ax, bx
mov [si], ax
jc l1
inc si
inc si
mov [si], dx
int 3
l1: inc dx
    inc si
    inc si
    mov [si], dx
    int 3
code ends
end start
```

OUTPUT :

075A : 3000 B8. AB 06. CD 00.FD 50. AE

- G= 0000

AX = 7CA8 BX = AEFD CX = 0023 DX = 0001 SP = 0600

BP = 0000 SI = 3006 DI = 0000 DS = 075A ES = 075A

SS = 0769 CS = 076A IP = 0022

```
C:\8086>DEBUG ADD16.EXE
-G=0000

AX=56B8  BX=5000  CX=0023  DX=0000  SP=0000  BP=0000  SI=3006  DI=0000
DS=075A  ES=075A  SS=0769  CS=076A  IP=001C   NV UP EI PL NZ NA PE NC
076A:001C CC            INT     3
-E
-E3000
075A:3000  B8.AB   06.CD   00.FD   50.AE

-G=0000

AX=7CA8  BX=AEFD  CX=0023  DX=0001  SP=0000  BP=0000  SI=3006  DI=0000
DS=075A  ES=075A  SS=0769  CS=076A  IP=0022   NV UP EI PL NZ NA PE CY
076A:0022 CC            INT     3
```

5) 16 bit subtraction :

```
code segment
assume cs:code
Start :
    mov ax, 0000h
    mov bx, ax
    mov dx, ax
    mov si, 3000h
    mov ax, [si]
    inc si
    inc si
    mov bx, [si]
    inc si
    inc si
    Sub ax, bx
    mov [si], ax
```

```
    jc l1
    inc si
    inc si
    mov [si], dx
    int 3
l1: inc dx
    inc si
    inc si
    mov [si], dx
    int 3
code ends
end start
```

OUTPUT :

075A :3000 B8. AE 06. FD 00. CD 50. AB

-G=0000

AX = 51E1 BX = ABCD CX = 0023 DX = 0000 SP = 0000 BP = 0000

SI = 3006 DI = 0000 DS = 075A ES = 075A SS = 0769 CS = 076A

IP = 001C

```
C:\8086>DEBUG SUB16.EXE
-G=0000

AX=B6B8  BX=5000  CX=0023  DX=0001  SP=0000  BP=0000  SI=3006  DI=0000
DS=075A  ES=075A  SS=0769  CS=076A  IP=0022   NV UP EI PL NZ NA PE CY
076A:0022 CC            INT     3
-E3000
075A:3000  B8.AE   06.FD   00.CD   50.AB

-G=0000

AX=51E1  BX=ABCD  CX=0023  DX=0000  SP=0000  BP=0000  SI=3006  DI=0000
DS=075A  ES=075A  SS=0769  CS=076A  IP=001C   NV UP EI PL NZ NA PE NC
076A:001C CC            INT     3
-
```

4) 16 bit multiplication :

```
code segment
assume cs : code
start :
    mov si, 7000h
    mov ax, [si]
    add si, 02h
    mov bx, [si]
    mul bx
    add si, 02h
    mov [si], ax
    add si, 02h
    mov [si], dx
    int 3
code ends
end start
```

OUTPUT :

```
075A:7000  10.AB  F7.12  46.CD  06.34
-G=0000

AX = AEEF  BX = 34CD  CX = 0017  DX = 03D9  SP=0000  BP=0000
SI = 7006  DI = 0000  DS = 075A  ES = 075A  SS = 0769
CS = 076A  IP = 0016
```

o/p verified

```
C:\8086>debug mul16.exe
-G=0000

AX=EE60  BX=0646  CX=0017  DX=060D  SP=0000  BP=0000  SI=7006  DI=0000
DS=075A  ES=075A  SS=0769  CS=076A  IP=0016   NV UP EI PL NZ NA PE NC
076A:0016 CC            INT     3
-E7000
075A:7000  10.AB   F7.12   46.CD   06.34

-G=0000

AX=AEEF  BX=34CD  CX=0017  DX=03D9  SP=0000  BP=0000  SI=7006  DI=0000
DS=075A  ES=075A  SS=0769  CS=076A  IP=0016   NV UP EI PL NZ NA PE NC
076A:0016 CC            INT     3
```